

Complete Git and GitHub Notes (Beginner to Advanced)

What is Git?

Git is a **distributed version control system** used to track changes in source code during software development. It allows multiple people to work on a project without interfering with each other.

What is GitHub?

GitHub is a **web-based platform** built on Git that provides hosting for software development and version control using Git. It allows collaboration, pull requests, code review, and project management.

Basic Git Setup

Install Git:

Download: <https://git-scm.com/downloads>

Configure Git:

```
git config --global user.name "Your Name"
```

```
git config --global user.email "your@email.com"
```

```
git config --global core.editor code # sets VS Code as editor
```

Check Configuration:

```
git config --list
```

Git Basics

Initialize a Git Repository:

```
git init
```

Creates a .git directory in your project folder to track changes.

Check Git Status:

```
git status
```

Add Files to Staging Area:

```
git add filename.txt    # single file
```

```
git add .               # all files
```

Commit Changes:

```
git commit -m "Your commit message"
```

View Commit History:

```
git log
```

View Logs in One Line:

```
git log --oneline
```

Rename Files:

```
git mv oldname.txt newname.txt
```

Remove Files:

```
git rm filename.txt
```

View Differences:

```
git diff
```

 Branching and Merging**Create a Branch:**

```
git branch new-branch
```

Switch Branch:

```
git checkout new-branch
```

Create and Switch:

```
git checkout -b new-branch
```

Merge Branches:

```
git checkout main
```

```
git merge new-branch
```

Delete a Branch:

```
git branch -d branch-name
```

Remote Repositories (GitHub)

Connect to Remote Repo:

```
git remote add origin https://github.com/username/repo.git
```

Push Code:

```
git push -u origin main # first time
```

Pull Changes:

```
git pull origin main
```

Clone a Repository:

```
git clone https://github.com/username/repo.git
```

View Remotes:

```
git remote -v
```

Advanced Git Concepts

Stash Changes (Temporary Save):

```
git stash
```

Apply Stash:

```
git stash apply
```

List All Stashes:

```
git stash list
```

Revert a Commit:

```
git revert commitID
```

Reset to Previous Commit:

```
git reset --hard commitID
```

Tagging Versions:

```
git tag v1.0
```

Show Tags:

```
git tag
```

Push Tags:

```
git push origin v1.0
```

Git Tools and Workflows

.gitignore File:

Used to ignore files/folders from being tracked.

Example .gitignore

```
node_modules/
```

```
*.log
```

```
.env
```

GitHub Flow:

1. Create branch
2. Make changes
3. Push to GitHub
4. Open Pull Request
5. Review & merge

Fork and Pull Model:

1. Fork repo
 2. Clone to local
 3. Push to your GitHub
 4. Create Pull Request to original repo
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🌟 Useful GitHub Features

- Issues: Track bugs/features
 - Projects: Kanban-style task board
 - Actions: CI/CD automation
 - GitHub Pages: Host static websites
 - Discussions: Community support and threads
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📅 Real-World Git Workflow Example

`git checkout -b feature/contact-form`

Make changes

`git add .`

`git commit -m "Add contact form"`

`git push origin feature/contact-form`

Go to GitHub → Open Pull Request → Review → Merge to main.

⚖️ Best Practices

- Use meaningful commit messages
 - Commit often
 - Pull before pushing
 - Use branches for features/bugs
 - Keep main/master stable
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Let me know if you'd like this as a **PDF**, **Notion page**, or if you want a **Git cheat sheet** or **GitHub Actions guide** next.